

# Aditya Kumar Gupta

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## SUMMARY

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Final-year Information Science student with strong proficiency in Python, SQL, and AI/ML fundamentals. Experienced in building end-to-end AI systems, including an LLM-based self-healing validation system and a crop yield prediction system. Skilled in machine learning and backend development using FastAPI and REST APIs. Hands-on experience in designing scalable pipelines, API integrations, and improving system reliability. Seeking entry-level roles in AI/ML, Data Science, or Backend development.

## WORK EXPERIENCE

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**Android Developer Intern — MindMatrix.io** Feb 2026 – Present

- Developed AI-powered applications integrating Generative AI APIs for real-time interactions.
- Designed scalable backend APIs enabling seamless communication between applications and AI services.
- Optimized system performance and response handling, improving reliability and reducing intervention.

## PROJECTS

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**AI Self-Healing Validation System**—Python, FastAPI, LLM, LangGraph [GitHub](#) — [Demo](#)

- Built an AI-driven system to detect and correct incorrect outputs using LLM-based validation pipelines.
- Designed a multi-step workflow (validation → correction → re-evaluation) to improve response reliability.
- Integrated LLM APIs with backend services using FastAPI for automated issue detection and resolution.
- Reduced manual intervention by ~30% by implementing iterative self-healing mechanisms.

**Smart Crop Yield Prediction System** —Python, FastAPI, Scikit-Learn [GitHub](#) — [Demo](#)

- Built a FastAPI-based REST API to process multi-variate agricultural inputs for crop yield prediction.
- Developed a Random Forest model trained on 43,000+ data points across crops and regions.
- Designed preprocessing pipelines using ColumnTransformer with scaling, encoding, and data handling.
- Improved decision support by delivering data-driven insights through an interactive dashboard interface.

## EDUCATION

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|----------------|---------------------------------------------------------------|-----------------|
| 2022 - present | B.E at Sir M. Visvesvaraya Institute of Technology, Bengaluru | (GPA: 8.2/10.0) |
| 2019 - 2021    | Intermediate(CBSE) at Holy Mission High School, Samastipur    | (Percent: 70.8) |
| 2018 - 2019    | Matriculation(CBSE) at St. Xaviers High School, Samastipur    | (Percent: 77)   |

## TECHNICAL SKILLS

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|--------------------------------------|------------------------------------------------------------------------------------------------------------|
| Languages:                           | Python, JavaScript, SQL (MySQL), HTML, CSS                                                                 |
| Backend & APIs:                      | FastAPI, RESTful APIs, Uvicorn, Pydantic, JWT Auth                                                         |
| Machine Learning & Data:             | Scikit-learn (Random Forest), Pandas, NumPy, Feature Engineering, Model Evaluation (RMSE, R <sup>2</sup> ) |
| AI & LLMs :                          | LLM APIs (Groq), LangGraph, AI Agent Systems                                                               |
| Data Visualization & Core Concepts : | Chart.js, Streamlit, OOP, DBMS, SQLite, OS, CN, DSA,                                                       |

## CERTIFICATIONS / VIRTUAL EXPERIENCE PROGRAMS

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**JPMorgan Chase & Co. – Quantitative Research Virtual Experience (Forage)** Oct 2025

- Analyzed loan data to estimate default probabilities using quantitative research methods.